



Gas and Liquids in Intercellular Spaces of Maize Roots

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Oils are spontaneously absorbed by gas-filled intercellular spaces (IS) in maize root cortex. The network of these spaces in living root sections was imaged by confocal laser scanning microscopy using a fluorescent solution of Nile red in oil. The gas volume fraction (GVF) of root segments was quantified by the increase in weight (differentiated zones) or tissue density (2–3 mm root tips) due to complete vacuum infiltration. Cooling to 6 °C or inhibition of oxidative phosphorylation diminished the GVF of root tips but did not significantly affect the GVF of differentiated root zones. The threshold pressure difference for measurable infiltration of isolated root segments is lower (10 to 15 kPa) than the threshold for infiltration of comparable zones of attached roots or of detached roots with the cut surface sealed (> 60 kPa). In the absence of an open cut, pressure-driven infiltration of the root cortex is accelerated by microscopic fissures within the epidermal/hypodermal barrier. The GVF of the root cortex was reduced after transferring roots from sugar solutions (0.1 to 0.3 M) to water. This points to efficient water transport from the medium to sugar-containing cortical cell walls through epidermal and hypodermal protoplasts. When 2-cm-long primary roots were vacuum infiltrated *in situ* and then allowed to grow on aerated mineral medium for a further 5 d, cortical IS of the originally infiltrated root bases remained filled with liquid but the subsequently grown apical root zones had a normal GVF.

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INTRODUCTION

Gas-filled intercellular spaces (IS) are generally present in extrafascicular plant parenchyma tissues (Sifton, 1945, 1957; Wolley, 1983; Passioura, 1988; Raven, 1996). The product of the oxygen diffusion coefficient and oxygen concentration is higher in air than in air-saturated water phases (cell walls, protoplasts) by a factor of about 4×10^5 . Therefore, interconnected gas-filled IS are the predominant pathways of oxygen diffusion through plant organs, even where their volume fraction is small (Stiles, 1960). In plant roots growing in an hypoxic environment, the oxygen conducting function of IS is a prerequisite for respiration of the tissues and creation of an appropriate redox potential in the rhizosphere (Jackson and Armstrong, 1999).

In water-saturated roots, cortical IS are usually gas-filled. It has been suggested by several authors (Woolley, 1983; Passioura, 1988; Raven, 1996) that IS of the root cortex might exclude water due to the hydrophobic character of their wall surfaces. Hydrophobic surface layers on the cell walls bordering IS are known in stems and leaves (Ursprung, 1924; Lewis, 1938; Häusermann, 1944). The hydrophobic character of the cell wall surface of root IS is more questionable. Staining of the cell wall surfaces of IS with lipophilic dyes and certain peculiarities of the cell wall ultrastructure have been detected in shoots but not in roots (Roland, 1978; Willison and Pearce, 1983). Partial water

filling of IS has been discussed as a possible reason for the spatial variance in the water content of *Pelargonium hortorum* roots found in an NMR-imaging study (Brown, Johnson and Kramer, 1988; Passioura, 1988). IS are immediately filled with liquid when living roots are denatured (Woolley, 1983). In a review article on radial water transport through roots, Passioura (1988) mentioned seeing slightly convex menisci in IS of the maize root and argued that tiny intercellular water channels might exist in hydrophilic edges where cell walls are separating from each other to form a narrow slit. It has been claimed that part of the IS of the root cortex is completely filled with sap containing K^+ ions in high concentration (Canny and Huang, 1992). However, these observations were made in rapidly frozen, not living, roots. There is considerable interest in the hypothesis of regulatory adjustment of O_2 diffusion resistance in root nodules of leguminosae by reversible liquid filling of IS (Tjepkema and Yocum, 1974; Witty, Skot and Revsbech, 1987; Parsons and Day, 1990; Purcell and Sinclair, 1994), although this mechanism has not been proven experimentally to be operational.

The presence of liquid in the cortical IS might be of great importance for the apoplastic route of radial volume fluxes of water through the extrastelar root tissues. Without such a phase, apoplastic water flow would be restricted to the cell wall matrix. The thin and dense cell wall material (Stokes' radius of excluded polymers about 3 nm) is many times more resistant to volume flow of water than are protoplasts. A very small volume fraction of liquid-filled IS can cause a

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dramatic reduction of the hydraulic resistance of parenchyma tissue (Michael *et al.*, 1997. Reports on efficient apoplastic water transport through the root cortex (e.g. Mees and Weatherley, 1957; Miller, 1985; Steudle and Boyer, 1985; Westgate and Steudle, 1985; Fiscus, 1986; Shone and Clarkson, 1988; Zhu and Steudle, 1991; Magnani, Centritto and Grace, 1996) may be due to a pathway through IS. However, in experimental arrangements it is difficult to exclude artificial infiltration. It has never been shown unequivocally that there is liquid in IS of an unperturbed living root system.

The present study deals with the stability of the gas filled volume fraction (GVF) in young maize roots. The lipophilic character of the cell walls bordering gas-filled IS is demonstrated by oil-absorption experiments, and the IS network is visualized by a fluorescent oil. Further results refer to osmotically-driven infiltration and to threshold values that have to be exceeded for pressure-driven infiltration.

MATERIALS AND METHODS

Seedling plants, root segments and detached roots with sealed cut surface

Maize caryopses (*Zea mays* L., 'Galina') were soaked in flowing tap water overnight and germinated for 2 d in the dark at 25 °C on wet cellulose sheets in a moist chamber. About 100 seedlings were placed on a polypropylene sieve with 4 mm-pores in a culture vessel with 6 l of aerated mineral-rich tap water (K^+ , Ca^{2+} 1–3 mM, Mg^{2+} > 0.2 mM) containing 100 mg l⁻¹ Penicillin G (Sigma, USA). They were grown for a further 5 d in a growth chamber with a daily light period of 12 h (PAR 300–400 μ mol m⁻² s⁻¹, 25 °C). Root segments were cut using a sharp razor blade.

To obtain a hydraulically tight seal at the cut surfaces, the ends (about 5 mm) of detached primary roots (length 8–10 cm) were covered with approx. 50 mg of a rapidly-hardening hydrophilic cap. The sealing material was freshly prepared by mixing 6 g gypsum with 10 ml warm 1 % agar solution (40 °C). During hardening of the seal (15 min) most of the root was submerged in tap water. The hardened seals were each covered by thin film of a silicone grease (Silikonfett NP 14, Nünchritzer Silikone, Nünchritz, Germany) and aluminium foil.

Infiltration procedures based on pressure changes

For low-pressure infiltration of root segments and short detached roots, a 0.5 l suction flask was used. Plant materials were kept submerged by fastening them to a rod or sample holder. They could be moved from the water to the decompressed air and *vice versa*. For this purpose, a thin, movable and greased cylindrical rod was fixed within the bored rubber stopper of the suction flask. The lower end of the rod carried a cylindrical holder for the sample (a bundle of roots or segments).

Over-pressure treatment of all materials investigated and low-pressure treatment of intact seedlings and long detached roots were carried out in a thick-walled 2 l glass conical flask. It was connected to the vacuum pump or compressed

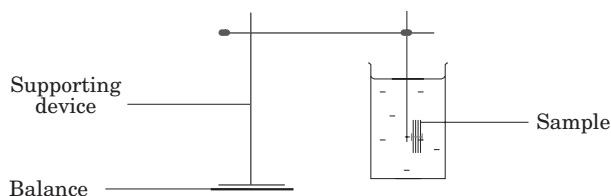


FIG. 1. Supporting device for weighing submerged samples.

air by a bored screw cap (diameter 3 cm). The vessel was kept in sand up to the level of the screw cap and almost filled with water before application of low or high pressure. Complete submersion of plant material in the flask was carried out, if necessary, by fixing it to a rod (short detached roots or root segments) or by keeping the roots under water by means of a water-saturated sponge pushed into the neck of the flask (long detached roots or intact seedling plants).

Oil treatment of intact seedlings

Plants were placed on perforated plates keeping the caryopses at a defined level within the cylindrical vessels. The vessels were filled with a commercial flax oil to a chosen height (sufficient to submerge either roots only, roots with part of the shoot, or the whole plant) and incubated in a water-saturated atmosphere within a glass chamber at room temperature.

Determination of the gas volume fraction (GVF)

The GVF of differentiated root zones was determined by measuring the change in weight when IS gas was replaced by water. In the case of experiments with intact plants, 3-cm-long segments, taken 2–8 cm from the root tip, were analysed. To avoid errors due to evaporation and non-reproducible surface films, the sample remained submerged during weighing (Fig. 1). A thin band of a commercial plastic and hydrophobic putty (Molto Fugenband weiß, Molto GmbH, Löhnberg, Germany) was used to bind the samples (bundled segments cut from at least 15 to 20 roots) and to stick them to a plate at the lower end of a cylindrical rod. This rod was fixed on a supporting device by a screw. The whole device was weighed on an electronic balance while the rod with the sample dipped in a water bath. After the tare had been ascertained, the sample was infiltrated (2.1 kPa, 10 min) as described above. The rod with the infiltrated sample was refixed to the supporting device on the balance, taking care that it dipped into the water bath to exactly the same (marked) depth as at the first weighing. Control experiments had shown that the second weight was not increased further by repeating the infiltration procedure. The GVF is the weight increase by infiltration as a percentage of the fresh weight of the infiltrated sample.

For determination of GVF in 2 to 3-mm-long root tips, which have a tissue density higher than water, the GVF could be determined more conveniently and accurately using tissue densities measured before and after complete infiltration. Twenty root tips were placed in a vessel with 20 ml water, and a solution of Ficoll (Pharmacia, Sweden, 400 kDa, 200 g l⁻¹) added to this volume through a displacement burette until half of the root tips exhibited

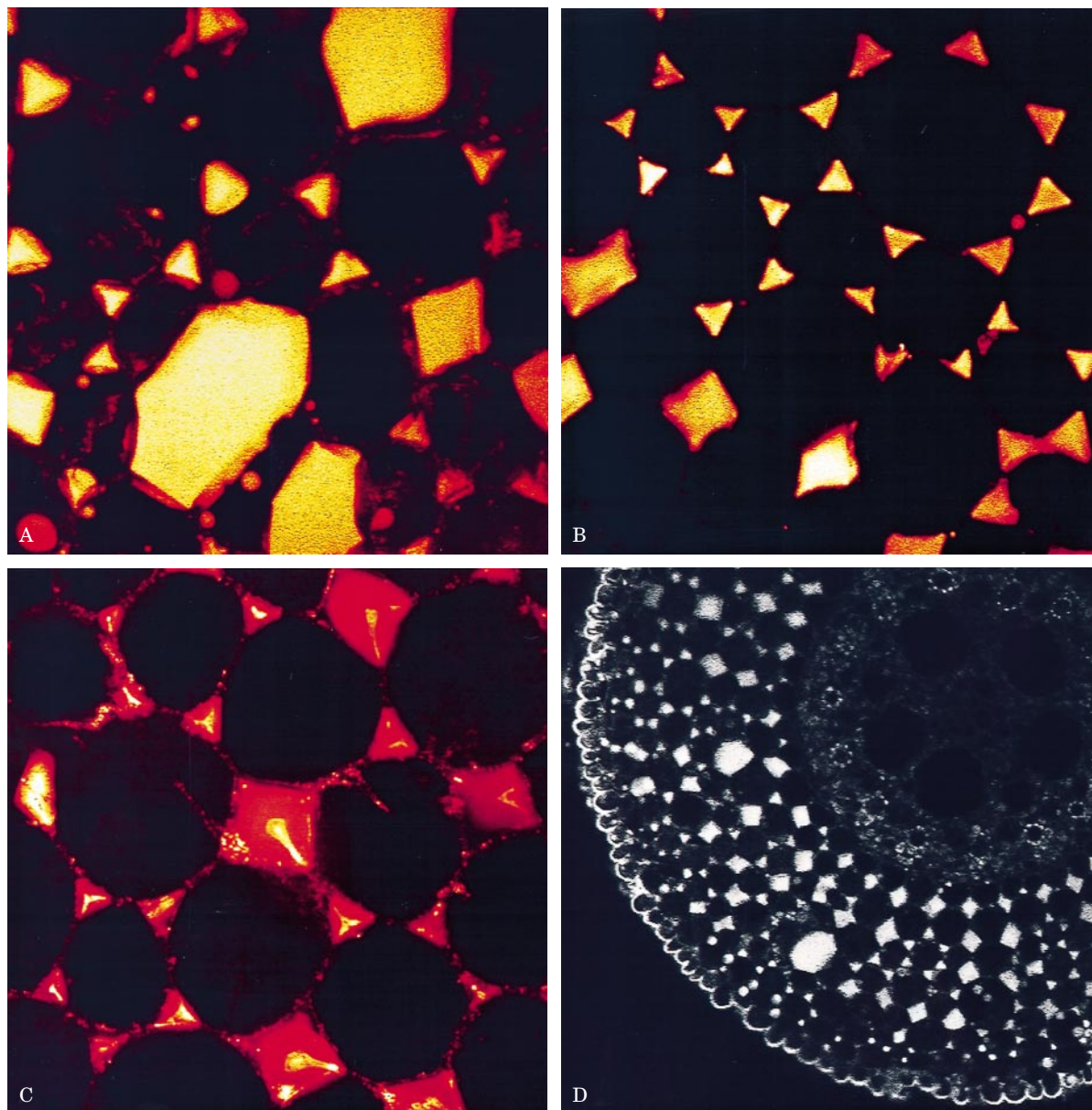


FIG. 2. Oil-loaded intercellular spaces in maize root cortex. Detached roots of 6-d-old maize seedlings were infiltrated in a solution of Nile red in thistle oil and sectioned 2.5 cm behind the tip. Fluorescence of a living 0.5 mm thick cross section was imaged by confocal laser scanning microscopy (excitation 510 nm, detection > 590 nm, focal plane approx. 20 μm below the section surface). A–C, Root cortex, 125 μm $\times$ 125 μm , increasing fluorescence intensity given by false colours (black < red < orange < yellow < white); D, root section, 500 μm $\times$ 500 μm , fluorescence intensity given as brightness.

positive buoyancy. The concentration and density of the isopycnic Ficoll solution was determined from the added volume by a calibration curve. The procedure was repeated with the same sample after an infiltrating treatment (10 min 2.3 kPa), to obtain the mean tissue density of the completely infiltrated tissue (σ_2). GVF is $(\sigma_2 - \sigma_1) / \sigma_2$.

Confocal laser scanning microscopy (CLSM)

A confocal laser scanning microscope of Leica Laser-technik GmbH, Germany, was used. Roots were detached, briefly blotted on cellulose sheet and infiltrated in a solution

of Nile red in thistle oil (1 mg l^{-1}) by 10 min application of low pressure (2.1 kPa). CLSM- fluorescence images of fresh hand sections (thickness about 0.5 mm, kept in water under a cover glass) were recorded approx. 30 min after infiltration.

RESULTS AND DISCUSSION

The network of gas-filled IS fills spontaneously with hydrophobic liquids

Root segments float on a water surface as long as they are turgescient (at least several days). All root segments sank to the bottom within 2 h after being blotted and placed

TABLE 1. Entry of thistle oil into roots of intact seedlings of *Zea mays*

Oil level	Gas volume fraction (GVF) and appearance of the roots after 20 h
Only roots submerged	20% not infiltrated, GVF 10.4% 45% incompletely infiltrated (partly glassy, mean GVF 3.2%) 35% completely infiltrated
Shoot partly submerged	Sample completely infiltrated
Shoot completely submerged	Sample completely infiltrated

5-d-old seedling were briefly blotted on cellulose pulp and placed on a perforated plate within vessels which were filled with thistle oil at different heights. The gas volume fraction (GVF) of the differentiated zones (2–8 cm) of infiltrated (more or less glassy) roots and non-infiltrated roots was determined separately.

in an oil mixture with the density of water (prepared from the high-density silicone oil AP 100, Fa. Wacker, München, Germany, and commercial flax oil). Roots infiltrated by the oil mixture remained turgid. The density of completely water-infiltrated root segments, as determined by the concentration of an isopycnic Ficoll solution, was 1.0143 g ml^{-1} . A simple calculation shows, that, at the instant of sinking in an oil with the density of water (0.997 g l^{-1}), the gas volume fraction (GVF) is about 1.8%, approx. one fifth of the original value (see below).

Fatty oils of relatively low viscosity (thistle oil, flax oil) spontaneously entered the roots of intact seedlings through the shoot (Table 1). All of the roots investigated were completely infiltrated when at least part of the leaf was submerged in these oils. When only roots were submerged, oil entered the root IS only in part of the plants and part of the oil-containing roots had a residual GVF. We assume that oil crossed the epidermis in part of the roots through small fissures created by handling of plants and blotting the roots.

After infiltration of the cortex with a solution of Nile red in thistle oil, the network of IS in the root cortex is clearly revealed by confocal laser scanning microscopy (CLSM). The chosen fluorescent dye is sparsely soluble in water and accumulates slowly in cellular lipoidal phases. When oil infiltration was accelerated by low-pressure treatment and images were recorded immediately after the infiltration step, the fluorescence was concentrated in the cortical IS (Fig. 2). With time, the fluorescence of the organelles within the cortical protoplasts and the fluorescence of the stelar tissue increased. All detectable IS were oil-filled. The CLSM image of the infiltrated maize root cortex shows many shizogenic and a few lysogenic IS. At certain focal planes, the individual oil-loaded IS appear to be interconnected in all directions (Fig. 2C). The IS close to the epidermis are smaller than those in the centre of the root cortex (Fig. 2D).

The GVF in differentiated root segments is not sensitive to low temperature and inhibitors

GVF was determined by measuring the increase in tissue

density or weight due to complete low-pressure infiltration. This was done generally with isolated root segments (differentiated zone or tip region). Previous experimental treatments were carried out with intact plants (Table 2) or isolated segments (Table 3). Local infiltration cannot be avoided in small zones close to the cut, where the cells are denatured. This is a minor error when the GVF of 2-cm-long root segments is considered, but is certainly more significant in the case of short root tips.

The GVF of the more differentiated root tissues was especially large when roots had been grown in unaerated nutrient solution. It increased with increasing distance from the tip but was zero after freezing and thawing (Tables 2 and 3). This confirms previous findings (Meiri and Anderson, 1970; Woolley, 1983; Van Nordwijk and Brouwer, 1988). The GVF of the differentiated cortex was not affected when the root medium of seedling plants was kept for several hours without aeration or when transpiration was inhibited by enclosing leaves in a polyethylene bag (Table 2). Nor did cooling have a significant effect on the GVF of the differentiated cortex, irrespective of whether intact seedlings or isolated segments were treated (Tables 2 and 3). A solution of 0.2 mM KCN and 0.1 mM dinitrophenol had no rapid effect on the GVF of isolated root segments 2 to 8 cm behind the tip (Table 3). The root IS remain gas-filled even in the face of severely inhibited ATP production. The same conclusion was drawn by Webb and Armstrong (1983) from the absence of any effect of CN^- and salicylhydroxamic acid on the oxygen conducting function of the IS in pea roots.

Treatment of the root segments with a detergent solution (5 mM sodium cholate) for 2 h reduced the GVF to zero (Table 3). Turgor of detergent-treated roots was strongly reduced. The observed reduction of GVF may result from the detergent effect on the hydrophobic cell wall surfaces or might be a consequence of lipid membrane damage and cell death.

The GVF of root tips was reduced when seedlings were cooled *in situ* to 6°C (Table 2). An even stronger reduction was found when isolated tips were shaken in the cold medium (Table 3). Likewise, treatment of isolated root tips (2 to 3 mm long) with 0.1 mM 2,4-dinitrophenol was also followed by a reduction of the GVF. Reductions of GVF by low-pressure infiltration, cold or inhibitor treatment were irreversible (Table 3).

Results shown in Tables 2 and 3 suggest that the GVF is more sensitive to inhibitors and cold treatment in tissues close to the meristem than in more differentiated regions. Increased solubility of gases at low temperature and rapid changes in tissue structure due to a decreased growth rate may contribute to the GVF reduction in the tip region. A considerable part of the gas volume in IS remains stable during prolonged incubation of isolated root tips in liquid medium. This suggests that the deposition of hydrophobic materials on the eldest cell wall layer (residues of middle lamella) is an early event in the formation of IS. Gas (CO_2) supersaturation due to the high respiratory quotient of the metabolically active tissue close to the meristem (Ramshorn, 1959) and growth-mediated reduction in water potential can favour gas-filling in this state, and the gas-wall interface itself may contribute to a further reduction of wettability.

TABLE 2. Gas volume fraction (GVF) in segments* of roots of *Zea mays* seedlings, 7 d after soaking

Expt	Distance from the root tip (cm)	Growth medium	Treatment of intact plants	GVF (%)†
I	0–0.2	Aerated	None	0.83 ± 0.03
	1–4			6.1 ± 0.8
	4–7			9.8 ± 1.0
	7–10			11.5 ± 1.8
II	1–4	Unaerated	None	12.0/14.8
	4–7			20.8/19.8
	7–10			24.9/23.7
III	4–7	Aerated	None	9.5 ± 3.0
		3 h unaerated	None	11.5 ± 2.4
		9 h unaerated	None	11.4 ± 2.8
IV	4–7	Aerated	None	9.8
			Transpiration inhibited for 5 h	9.5
V	4–9	Aerated	None	10.0 ± 2.0
			5 h at 6 °C	9.6 ± 3.1
VI	0–0.2	Aerated	None	1.4
			2 h at 6 °C	1.1
VII	0–0.3	Aerated	None	1.8
			2 h at 6 °C	1.7
			20 h at 6 °C	1.3

* Roots were cut into segments for determination of GVF; † unique determinations (IV, VI, VII), two replicate measurements (II), or mean values ± s.d. from six (I) or three (III) replicates.

TABLE 3. Effect of different treatments on the gas volume fraction of isolated root segments of *Zea mays*

Expt	Distance from the tip	Treatment after isolation	Duration of treatment (h)	GVF (%)
I*	4–7	0.2 mM CaCl ₂ (control)	8	8.6 ± 0.5
		0.1 mM DNP	8	9.0 ± 2.7
		frozen/thawed		0
		5 mM sodium cholate	2	0
II	4–7	0.2 mM CaCl ₂ (control)	4	9.8/11.2
		6 °C	4	10.2/11.2
		0.2 mM KCN	4	10.1/11.0
III	4–7	Aerated in 0.2 mM CaCl ₂ after low-pressure infiltration	20	0/0.1
		Aerated in 0.1 M glucose after low-pressure infiltration	20	0/0
IV	0–0.3	0.2 mM CaCl ₂ (control)	2	1.48
		0.1 mM DNP	2	1.21
		0.1 mM DNP,	2	
		subsequent incubation	4	1.23
		without DNP		
		6 °C	4	0.36
		6 °C, subsequently at 24 °C	4	
			2	0.33

* Mean values ± s.d., three replicates.

Segments were cut from the primary root 5 d after soaking and shaken for the time indicated in the respective solution. Unless indicated otherwise the temperature was 22 to 24 °C.

Capillary depression of water in the hydrophobic IS results in a measurable threshold value for pressure driven infiltration

During normal low-pressure infiltration (transient reduction of absolute liquid pressure to a few kPa) the IS are

filled immediately upon recompression, when an inward directed transient pressure difference is created between the external liquid and the IS. The process is very rapid, not only in isolated root segments but also in roots of intact seedlings, as indicated by the immediate change from opalescence to translucence.

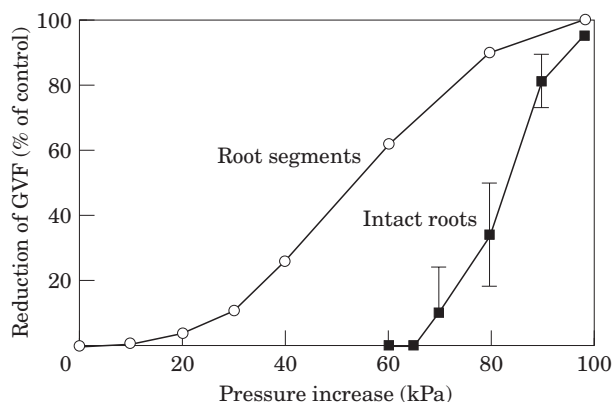


FIG. 3. Infiltration pressure threshold of submerged root segments and roots of intact seedlings of *Zea mays*. Bundles of 3-cm-long root segments (4–7 cm behind the tip, cut from 6-d-old plants) or 5-d-old intact seedlings were completely submerged and subjected to reduced pressures for 10 min to allow for the indicated pressure increase from below-atmospheric to atmospheric pressure. ○, Isolated root segments (mean values, $n = 3$); ■, roots of intact seedlings ($n = 5$ or 6). Control values of GVF (pressure unchanged) were $7.1 \pm 0.3\%$ for isolated segments and $10.4 \pm 2.0\%$ for roots of intact seedlings.

Different values of below-atmospheric pressure (absolute pressure between 2.3 to 91 kPa) were applied to the medium of submerged root segments and, after restoration of atmospheric pressure (approx. 101.3 kPa) the GVF was determined. A threshold pressure difference of about 15 kPa had to be exceeded to obtain a detectable decrease in the GVF. The pressure difference had to be about 50 kPa to reduce GVF by 50% (Fig. 3). The threshold pressure difference was 10 to 15 kPa irrespective of whether the pressure was increased to atmospheric pressure from a lower value (Fig. 3) or from atmospheric pressure to a higher value (data not shown).

The reduction of absolute pressure to 2.1 kPa does not seem to reduce the threshold pressure difference. No significant infiltration was found when the absolute pressure on submerged root segments was increased from 2.1 kPa to 12 kPa, the roots were then lifted to the gaseous space in the suction flask and the final increase to atmospheric pressure was carried out when the root segments were above the water surface (Table 4). Thus, the infiltration threshold of root segments was not removed by gas decompression. If adsorbed gas molecules contribute to the threshold, a higher vacuum is needed for desorption of these molecules. Threshold values can be calculated roughly as capillary depression (Δp) in a cylindrical hydrophobic pore with radius r

$$\Delta p = -2\sigma \cos \alpha / r \quad (1)$$

where σ is the surface tension of water (7.2×10^{-3} Pa m) and α is the contact angle. At the maximum contact angle of water (180°), capillaries with a radius (r) of 10 μm have a threshold (Δp) of 14.4 kPa, similar to the measured value. As the larger IS of maize roots can be about 20 μm in diameter (Fig. 1), the result of the above calculation suggests that $\cos \alpha$ is not very far from -1 , or, in other words, the cell wall surface of gas-filled IS is highly hydrophobic.

TABLE 4. Gas volume fraction of isolated root segments of *Zea mays* seedlings subjected to a pressure increase of 10 kPa at different absolute pressures

Variant	Final GVF (%)
I* Control, untreated	$8.7 \pm 1.5^\dagger$
II† 2.1 kPa (S) $\Rightarrow$ 101 kPa (S)	0
III† 2.1 kPa (S) $\Rightarrow$ 2.1 kPa (E) $\Rightarrow$ 101 kPa (E)	8.5
IV† 2.1 kPa (S) $\Rightarrow$ 12 kPa (S) $\Rightarrow$ 12 kPa (E) $\Rightarrow$ 101 kPa (E)	8.6
V† 91 kPa (S) $\Rightarrow$ 101 kPa (S)	8.3

3-cm-long root segments were prepared in a distance of 1–4 cm behind the tip and subjected for 10 min each to the given absolute pressures within a suction flask when either submerged (S) or emerged (E) (see Materials and Methods).

* Mean values of four replicates $\pm$ s.d.; † mean values of two replicates.

When water is driven into gas-containing IS by a pressure drop between the liquid medium and the IS, the gas within the IS is compressed but not immediately removed. The GVF does not fall to zero until the compressed gas within the IS is completely dissolved. Strong compression (by 0.1 MPa or more) of the IS gas is followed by its dissolution since the solubility of atmospheric gases in water is proportional to the pressure. Root segments were indeed completely infiltrated by over-pressure infiltration of sufficient strength (data not shown).

When isolated segments or intact plants are submerged within a large volume of degassed water, an inward pressure gradient between the IS and the liquid medium develops by dissolution of IS gas. After submerging isolated root segments (differentiated zone) or intact seedlings in degassed water for 2 h, the GVF was not significantly different from zero (two experiments each with 5-d-old plants). Again, infiltration of roots in degassed water was not observed when the shoots of seedling plants were in contact with the gaseous atmosphere.

The infiltration threshold is higher for intact roots than for isolated segments

When the shoot of seedlings emerged only a few millimeters above the water surface, roots were not infiltrated by low- or high-pressure treatments. Obviously, in this case, the gas pressure in the cortical IS equilibrates too rapidly with the ambient gas pressure for significant volume flow of water to be possible. The threshold for root infiltration of completely submerged plants or for roots with a sealed cut surface was found to be significantly higher (65–70 kPa) than that of isolated root segments (Fig. 3). The higher threshold value may be explained by the inverse relation between capillary depression and capillary width [eqn (1)]. In the absence of a cut surface, water has to cross the epithelial tissues (epidermis and hypodermis) before the cortical IS are filled. Close to the root surface, the IS are very small

TABLE 5. *Infiltrating effect of over-pressure to attached roots of submerged seedlings of Zea mays*

Expt	Absolute pressure (kPa) (%)	Duration of treatment (min)	GVF*
I	101 (control)	10	10.5 ± 1.1
	181	10	8.4 ± 0.5
II	101 (control)	20	10.4 ± 0.8
	185	20	7.6 ± 0.4
III	101 (control)	30	10.2/9.4
	201	10	9.1/10.1
	201	30	5.6 ± 2.8

Intact seedlings were subjected to over-pressure for the indicated period before GVF was determined in the differentiated root zone, 3–8 cm behind the tip.

* Mean values of three replicates ± s.d., or two replicates.

and usually not connected directly with the external liquid (Fig. 2). To initiate infiltration, the gas phase has to be replaced by water in narrow IS, or water has to break through a hydrophobic polymer network at the cell wall surface.

When intact seedling plants were subjected to low pressure (< 30 kPa), emergence of gas bubbles at distinct sites of the root surface was regularly observed. As the soft epithelial tissues at the root surface are not supported mechanically against an outward directed pressure difference, the evacuation step itself may create or enlarge fissures within the epidermis and hypodermis. Low-pressure infiltration of roots without an open cut surface was generally rapid as indicated by rapid sinking (some seconds) after pressure increase. Reduction of the GVF at over-pressure infiltration of attached roots or detached roots with sealed cut surfaces required more time. Intact plants submerged for 10 min at an absolute pressure of 180 to 200 kPa showed no significant decrease in their GVF (Table 5). Even exposure for 30 min caused infiltration of only part of the roots. Samples with reduced GVF obtained by over-pressure application generally showed a heterogeneous pattern of infiltration.

Individual roots were either translucent or opalescent over their whole length. Obviously, over-pressure infiltration is a rapid process initiated at different times in individual roots.

Transient osmotic gradients cause infiltration

Equilibration of root segments or intact seedlings with solutions of μ -methylglucoside (0.1 to 0.3 M) had no marked effect on GVF. However, when the segments or seedlings were transferred from the sugar solutions to water, the GVF was markedly reduced (Table 6). After transfer of the roots with the sugar-loaded cortical free space (mainly cell walls) to water, there is a transient sugar concentration gradient between the cortical apoplast and the medium. Sugar diffusion out of the apoplastic free space of maize roots has a half life period of 2–3 min (Göring, Ehwald and Sammler, 1974; Fleischer and Ehwald, 1995). We assume that the observed volume flux into the cortical apoplast is osmotically driven and crosses the epidermis and hypodermis by the protoplasmic route. Reflection coefficients of sugars in cell walls are close to zero and the hydraulic conductivity of the cell wall pathway is extremely small (Michael *et al.*, 1997).

Osmotic and hydrostatic pressure differences drive volume fluxes through the epidermis and hypodermis by different pathways

Pressure can drive water through the epithelial layers by the same route as the osmotic force. However, in contrast to osmotically-driven flow, pressure-driven flow is not restricted to semi-permeable membrane-bound areas. We have argued (see above) that small fissures in the epidermis and hypodermis are the favoured pathway of pressure-driven infiltration in the absence of an open cut surface. If this is true, a reduction in osmotic potential of the medium cannot counteract pressure-driven infiltration efficiently. Indeed, we found that infiltration by a pressure increase from 2 to 101 kPa or from 101 kPa to 250 kPa was neither prevented nor significantly inhibited in PEG 6000 solutions

TABLE 6. *Infiltrating effect of a transient osmotic potential difference between the medium and the cortical free space of Zea mays roots*

Material	Preincubation	A	GVF (%) A + B
Isolated segments	20 h, 0.2 mM CaCl ₂	9.9 ± 1.0	
	20 h, 100 mM α -methylglucoside (0.27 MPa)	8.9	5.3 ± 1.3
	20 h, 175 mM α -methylglucoside (0.47 MPa)	9.2	4.3 ± 0.5
Intact seedlings	2 h, 0.2 mM CaCl ₂	12.5 ± 0.7	
	2 h, 200 mM α -methylglucoside (0.54 MPa)†	11.2	9.5 ± 0.6
	2 h, 300 mM α -methylglucoside (0.82 MPa)†	7.8	3.8 ± 1.1

6-d-old plants or their isolated root segments (2–8 cm behind the tip) were preincubated in different solutions and subsequently shaken in 0.2 mM CaCl₂ for the periods indicated. GVF was determined after preincubation (A) and after subsequent transfer to 0.2 mM CaCl₂ (A + B).

The osmotic potential of the root cell sap was close to –1.2 MPa.

* Mean value ± s.d., three replicates or individual measurements.

TABLE 7. Effect of PEG 6000 on pressure-driven infiltration of Zea mays roots

Material	Concentration, osmotic potential of PEG 6000 g l ⁻¹ kPa		Change of absolute pressure used for infiltration kPa	Final GVF* %
Intact seedling plants	0	0	2.3 → 101	0.3/0.7
Untreated	40	23	2.3 → 101	0.0/0.8 10.4 ± 0.5
Detached roots with sealed cuts	0	0	2.3 → 101	3.0
	100	110	2.3 → 101	2.3
	0	0	101 → 251	0.0/1.4
	175	280	101 → 251	4.8/3.0
	200	≈ 400	101 → 251	4.8/1.2
Untreated				9.6

Intact seedling plants (6 d after soaking) or intact detached roots with sealed cut ends (see Materials and Methods) were subjected to low pressure (2.3 kPa) for 10 min or to over-pressure for 60 min.

All solutions contained 0.2 mM CaCl₂.

* Mean value ± s.d., three replicates, or individual measurements.

of comparably low osmotic potential (Table 7), although this polymeric osmoticum does not cross maize root cell walls (Michael, 1999).

Infiltration of the IS is not reversed in submerged primary roots

If the cell wall surfaces in IS remained hydrophobic in the infiltrated state, this should favour the restoration of a gas phase in these spaces by liquid absorption. To check whether the seedling plant is able to drain the solution from infiltrated root IS, 2-d-old seedlings with a uniform root length (2 cm) were selected and their roots infiltrated. These and untreated control seedlings ($n = 20$) were grown for a further 5 d on aerated medium in a growth chamber (12 h, PAR 300–400 $\mu\text{mol m}^{-2} \text{s}^{-1}$). In control seedlings, the roots reached a mean length of 14–16 cm, some secondary roots appeared, and the primary leaves were well developed. Infiltration of the primary root prevented the further development of shoots and formation of secondary roots. The infiltrated primary roots, however, continued growth, although at a lower rate than the non-infiltrated roots of control plants (final length 6–8 cm). The IS of the originally infiltrated zone (2 cm at the root basis) remained completely filled with liquid (GVF close to zero) over the whole growth period. The GVF of younger root parts (developed after the infiltration treatment, 2–6 cm from the tip) was similar to that of control plants (9–10%). It is possible that in infiltrated IS the properties of the cell wall surface change so that water adheres. Further studies should focus on identification of the lipophilic materials in IS of the root cortex and their fate during liquid infiltration.

Consequences for water transport studies

We would like to stress that the GVF of the submerged primary maize root cortex is stable and rather independent of metabolism *in situ*, as it is based on a special cell wall structure which prevents wetting of the cell wall surface with water by capillary depression. This is certainly related to the

essential physiological function of the cortical IS as a gas conducting system for respiratory root metabolism. On the other hand, the hydrophobic character of the IS cell walls in the cortex of maize roots disappears at infiltration, and this has important consequences for water transport studies. Infiltration creates a low-resistant pathway for water through the root cortex in both an axial and radial direction. This, together with the demonstrated osmotically-driven volume fluxes, should allow determination of the ultrafiltration coefficient

$$\left(L_{\text{PD}} = \frac{J_v}{\Delta\pi} \right)$$

of the epithelium-like root surface tissues (epidermis, hypodermis).

Attached roots showed the highest resistance to infiltration. Significant osmotic or hydrostatic driving forces are necessary to wet their IS (Tables 5–7). Such driving forces are unlikely to exist in natural habitats but their infiltrating effects have to be thoroughly considered in laboratory experiments on radial water transport. Published arguments for the presence of a low-resistant hydraulic bypass through the cortical apoplast (see introduction) should be reconsidered critically in the light of this finding. A very small fraction of extracellular water can increase the hydraulic conductivity of a tissue dramatically (Michael *et al.*, 1997). There are several reasons for partial infiltration of detached roots used in laboratory experiments. It is not possible to avoid infiltration of IS close to a cut surface where some layers of cells are usually denatured. Local infiltration via microscopic wounds may result as a consequence of mechanical stress imposed by handling of the root. Pressure-driven infiltration has to be expected when applying pressure in chambers (Mees and Weatherley, 1957; Steudle and Boyer, 1985). A rapid increase in the xylem pressure above the equilibrium value (Steudle, 1994) might drive water into cortical IS. We have shown in this study that cortical IS became infiltrated when roots were subjected to osmotic changes or when they were detached and submerged in a medium which was not gas-saturated.

CLSM did not show oil/water menisci in edges of previously gas-filled IS (Fig. 2). Presumably, in the undisturbed root cortex, water is localized exclusively in cells (protoplasts and walls). If this is true, radial water transport through the cortex is mainly protoplasmic, since the cell wall matrix around protoplasts has a higher hydraulic resistance than protoplasts (Michael *et al.*, 1997).

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